KARP-RABIN fingerprints

How to quickly check files using fingerprints?
How to check for computer viruses?

Idea | Binary alphabet $\{0,1\}$

sequence $S = 01\,//\,0100110001\ldots//01$ $\quad n = |S|$

it can be seen as a very very large number $N_S \in [0..2^n - 1]$

sequence $S' = 01111\ldots$ $\quad n = |S'|$

It costs $O(n)$ to check equality

$F(S) = N_S \% p$ for a random prime $p \in [2..z]$ where $z \ll 2^n$

↑ user chooses it

Issue

$F(S) \neq F(S') \Rightarrow S \neq S'$ ✓

$F(S) = F(S') \begin{cases} \nearrow S = S' & \checkmark \\ \searrow S \neq S' & \leftarrow \text{1-sided error} \end{cases}$

undavoidable : what is the probability?

ERROR $k_1 \rightsquigarrow N_S$ $\quad k_1 \neq k_2 \wedge (k_1 \% p = k_2 \% p)$

$k_2 \rightsquigarrow N_{S'}$

Idea
- choose τ sufficiently large (see later on)
- choose uniformly and randomly a prime in $[2 .. \tau]$

Before starting the computation

p is a BAD prime if $k_1 \% p = k_2 \% p$

example: $k_1 = 37$, $k_2 = 40$

$p=3$ is BAD $p=5$ is OK

$$\Pr(\text{error}) = \Pr(k_1 \neq k_2 \wedge (k_1 \% p = k_2 \% p)) = \frac{\#\text{ BAD primes}}{\#\text{ primes in } [2..x]}$$

$$= \frac{?}{x / \ln x} \quad \leftarrow \text{density of primes} \qquad \leq \frac{n}{x / \ln x}$$

With high probability $= \frac{1}{n^c}$ $\quad c = \text{constant} > 1$

to fix x, it suffices to set $\frac{n}{x/\ln x} \leq \frac{1}{n^c}$

$\Rightarrow x \sim n^{c+1} \ln n$

Fact Any number x with n bits has $< n$ distinct primes dividing x

proof Factorization theorem

$$x = p_1^{e_1} p_2^{e_2} \cdots p_r^{e_r} \qquad 2025 = 3^4 \times 5^2$$

$$p_1^{e_1} \times p_2^{e_2}$$

$p_i \geq 2 \Rightarrow e_1 + e_2 + \cdots + e_r \leq n$

as otherwise $x \geq 2^{e_1+e_2+\cdots+e_r} > 2^n$ not possible

since $e_1 + e_2 + \cdots + e_r \leq n \Rightarrow r = \#$ distinct primes dividing $x \leq n$

□

p is BAD if it divides both k_1 and $k_2 \Rightarrow$ at most n BAD primes

What if $k_1 \% p = k_2 \% p$ but p does not divide k_1, k_2?

We can apply this same argument to $(k_1 - k_2) = k$

37 and 40 are not multiple of 3

but 40-37 is so.

Application : Rolling Hashing

https://github.com/lemire/rollinghashcpp

Text string ABRACADABRA

m=4

m=4

$F(BRAC) = B \cdot 131^3 + R \cdot 131^2 + A \cdot 131 + C \cdot 131^0 \ \% p$ ASCII

$F(RACA) = R \cdot 131^3 + A \cdot 131^2 + C \cdot 131 + A \cdot 131^0 \ \% p$

O(m) time

$$F(RACA) = (F(BRAC) - B \cdot 131^3) \cdot 131 + A \ \% p$$

↑

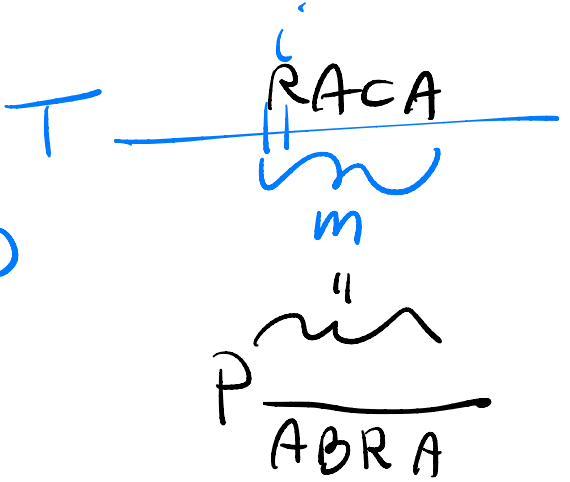
O(1) time ...

STRING MATCHING:

P = pattern string : ABRA m

T = text string = ABRACADABRA n

If P occurs in T ...



BASELINE : for $i = 1 \dots n-m+1$:
compare P with $T[i \dots i+m-1]$

$O(n \cdot m)$ time

KR - algorithm

$p = F(P)$

$t = F(T[1 \dots m])$

for $i = 1 \dots n-m+1$:

compare p with t

update t in $O(1)$ time so that $t = F(T[i+1 \dots i+m])$

$$X_i = \begin{cases} 1 & \text{error} \quad p = t \text{ but } P \neq T[i..i+m-1) \\ 0 & \text{o.w.} \end{cases}$$

$$\Pr[X_i = 1) = \frac{m}{\tau / \ln \tau}$$

$$X = \sum_{i=1}^{n-m+1} X_i > 0 \text{ iff there is an error at some point}$$

$$\Pr(\text{error KR-algorithm}) \text{ is } X \leq n \cdot \frac{m}{\tau / \ln \tau} \leq \frac{1}{n^c}$$

1) $m \leq n \Rightarrow mn \leq n^2$

2) $\tau \sim n^{c+2} \ln n$

MONTECARLO ALGORITHM

- running is guaranteed to be $O(n+m)$ time
- 1-Side error

LAS VEGAS ALGORITHM

- running time is expected
- no errors

KR-algorithm: LAS VEGAS

if $(p=t)$ then if $P = T[i..i+m-1]$ then RETURN Ok

Running time is $O(n+m+X\cdot m)$

Expected running time is $O(n+m+E[X]\cdot m)$

$E[X] \leq \frac{n^2}{\tilde{\iota}/\ell n \tilde{\iota}}$ $\cdot m$ $\rightarrow O(n)$